

The rectangular coordinate system in space

A third axis, and the two formulas — distance and midpoint — that everything else in the chapter is built on.

LESSON 1–3

3 × 40 MIN

GEOMETRY 11, §1

P1 · 3-D COORDINATES

LESSON CLOCK

14'

22'

30'

24'

24'

21'

The third axis	14 min
Plotting and reading points	22 min
The distance formula	30 min
The midpoint formula	24 min
The sphere	24 min
Practice and homework	21 min

LEARNING OBJECTIVES

- Plot a point from its three coordinates and read coordinates from a diagram.
- Use the distance formula in space.
- Use the midpoint formula in space.
- Find the equation of a sphere from its centre and radius.

TEXTBOOK

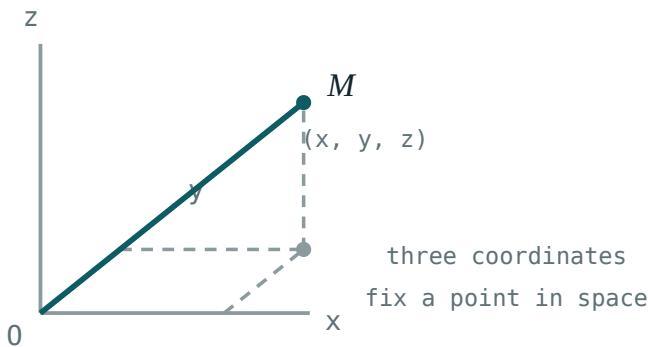
National	pp. 3–16
Cambridge	Core & Extended, pp. 318–324
Workbook	Exercise 1.1

01

Explanation

A third axis

Add an axis Oz perpendicular to both Ox and Oy . Every point of space then has exactly three coordinates (x, y, z) — how far along, how far across, how far up.



The point $(3, 4, 5)$ and the box that locates it. Read along, across, then up.

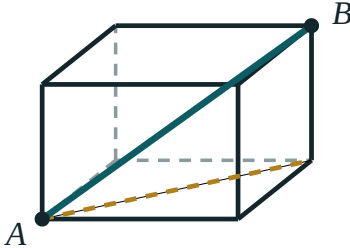
OBJECT	DESCRIPTION
the three axes	Ox, Oy, Oz
the three coordinate planes	$z = 0, y = 0, x = 0$
octants	8
a point on the x -axis	$(a, 0, 0)$
a point in the xy -plane	$(a, b, 0)$

RIGHT-HANDED CONVENTION

Point the fingers of the right hand along Ox and curl them towards Oy ; the thumb points along Oz . Every formula in this chapter assumes it.

Distance

Pythagoras, used twice — once in the base plane and once vertically:



Pythagoras
used twice

$$AB = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$$

The base diagonal first, then the vertical rise. One right triangle inside another.

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

From the origin it shortens to $OA = \sqrt{x^2 + y^2 + z^2}$. For $(3, 4, 12)$ that is $\sqrt{9 + 16 + 144} = 13$.

SQUARE THE DIFFERENCES, NOT THE COORDINATES

$A(1, 2, 3)$ and $B(4, 6, 15)$ give $\sqrt{9 + 16 + 144} = 13$ — from the **differences** 3, 4, 12. Squaring the coordinates themselves is the commonest slip.

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Each coordinate is averaged independently — the plane formula with one more entry.

More generally, the point dividing AB in the ratio $m : n$ from A is

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n}, \frac{nz_1 + mz_2}{m + n} \right)$$

and the midpoint is the case $m = n$.

The sphere

A sphere is the set of points a fixed distance from a centre. Writing that with the distance formula and squaring gives its equation directly:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

Centre $(2, -1, 3)$, radius 5: $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 25$.

GOING BACKWARDS

Given $x^2 + y^2 + z^2 - 4x + 6y - 12 = 0$, complete the square in each variable: $(x - 2)^2 + (y + 3)^2 + z^2 = 25$.

Centre $(2, -3, 0)$, radius 5.

02 Worked examples

EXAMPLE 1 Find the distance between $A(1, -2, 3)$ and $B(4, 2, 15)$.

① Differences 3, 4, 12.

② $9 + 16 + 144 = 169$

③ $\sqrt{169} = 13$

ANSWER

13

EXAMPLE 2 $A(2, -3, 5)$ and $M(4, 1, 0)$ is the midpoint of AB . Find B .

① $\frac{2+x}{2} = 4 \Rightarrow x = 6$

② $\frac{-3+y}{2} = 1 \Rightarrow y = 5$

③ $\frac{5+z}{2} = 0 \Rightarrow z = -5$

ANSWER

$B(6, 5, -5)$

EXAMPLE 3 Find the centre and radius of $x^2 + y^2 + z^2 - 6x + 4z - 3 = 0$.

① $(x - 3)^2 - 9$

② $(z + 2)^2 - 4$

③ $(x - 3)^2 + y^2 + (z + 2)^2 = 16$

④ Read off.

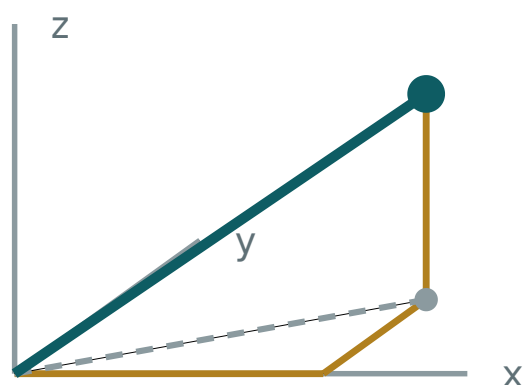
ANSWER

centre $(3, 0, -2)$, radius 4

03 Interactive model

Move the point and watch each coordinate and the distance from the origin update.

A point in space



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate system in space	Fazoviy koordinatalar sistemasi	Пространственная система координат
Ordinate / abscissa / applicate	Ordinata / abssissa / applikata	Ордината / абсцисса / аппликата
Octant	Oktant	Октант
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Distance between two points	Ikki nuqta orasidagi masofa	Расстояние между точками
Midpoint of a segment	Kesma o'rtasi	Середина отрезка
Sphere	Sfera	Сфера
Projection onto a plane	Tekislikka proyeksiya	Проекция на плоскость
Origin	Koordinatalar boshi	Начало координат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The distance from the origin to $(2, 3, 6)$ is:

The midpoint of $(0,0,0)$ and $(4, 6, 8)$ is:

A point in the xy -plane has:

How many octants?

$(x - 1)^2 + y^2 + (z + 2)^2 = 9$ has radius:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Distance from O to $(1, 2, 2)$

ANSWER 3

2. Distance from O to $(2, 3, 6)$

ANSWER 7

3. Distance from O to $(3, 4, 12)$

ANSWER 13

4. Midpoint of $(0,0,0)$ and $(2, 4, 6)$

ANSWER $(1, 2, 3)$

5. Midpoint of $(1,1,1)$ and $(3,5,7)$

ANSWER $(2, 3, 4)$

6. Radius of $x^2 + y^2 + z^2 = 36$

ANSWER 6

7. A point on the z -axis has form

ANSWER $(0, 0, c)$

Medium · the standard question

7 PROBLEMS

1. Distance $A(1,2,3)$ to $B(4,6,15)$

ANSWER 13

2. Distance $A(-1,0,2)$ to $B(3,4,2)$

ANSWER $4\sqrt{2}$

3. Midpoint of $(-2, 5, 1)$ and $(6, -1, 7)$

ANSWER $(2, 2, 4)$

4. Sphere, centre $(1,2,3)$ radius 4 — its equation

ANSWER $(x-1)^2 + (y-2)^2 + (z-3)^2 = 16$

5. Centre and radius of $x^2 + y^2 + z^2 - 4x = 0$

ANSWER $(2,0,0), 2$

6. $M(3,1,2)$ is the midpoint of $A(1,-1,0)B$. Find B

ANSWER $(5, 3, 4)$

7. Is $(1,2,2)$ on the sphere $x^2+y^2+z^2 = 9$?

ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Show $A(1,2,3)$, $B(3,4,5)$, $C(5,6,7)$ are collinear

ANSWER Equal steps of $(2,2,2)$

2. Find z so $(1, 2, z)$ is 7 from $(4, 6, 1)$

ANSWER $z = 1 \pm 2\sqrt{6}$

3. Point dividing $A(1,0,2)B(7,6,8)$ in ratio $1:2$

ANSWER $(3, 2, 4)$

4. Centre and radius of $x^2+y^2+z^2 -2x +4y -6z -11 = 0$

ANSWER $(1, -2, 3), 5$

5. Find the point equidistant from $(0,0,0)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$

ANSWER $(1, 1, 1)$

6. Show $A(0,0,0)$, $B(4,0,0)$, $C(4,3,0)$, $D(0,3,0)$ form a rectangle

ANSWER Opposite sides equal, diagonals equal 5

7. A cube has vertices $(0,0,0)$ and (a,a,a) . Find the space diagonal

ANSWER $a\sqrt{3}$

07 Homework

Homework — 6 tasks

Show the differences before squaring in every distance question.

- Find the distance between $A(2, -1, 4)$ and $B(5, 3, 16)$.
- Find the midpoint of $A(-3, 2, 7)$ and $B(5, -4, 1)$.
- $M(1, 0, 2)$ is the midpoint of AB with $A(-2, 3, 5)$. Find B .
- Write the equation of the sphere with centre $(-1, 2, 0)$ and radius 6.
- Find the centre and radius of $x^2 + y^2 + z^2 + 2x - 8y + 4z + 5 = 0$.

6. Show that $A(1,1,1)$, $B(3,3,3)$, $C(6,6,6)$ are collinear and find the ratio $AB : BC$.